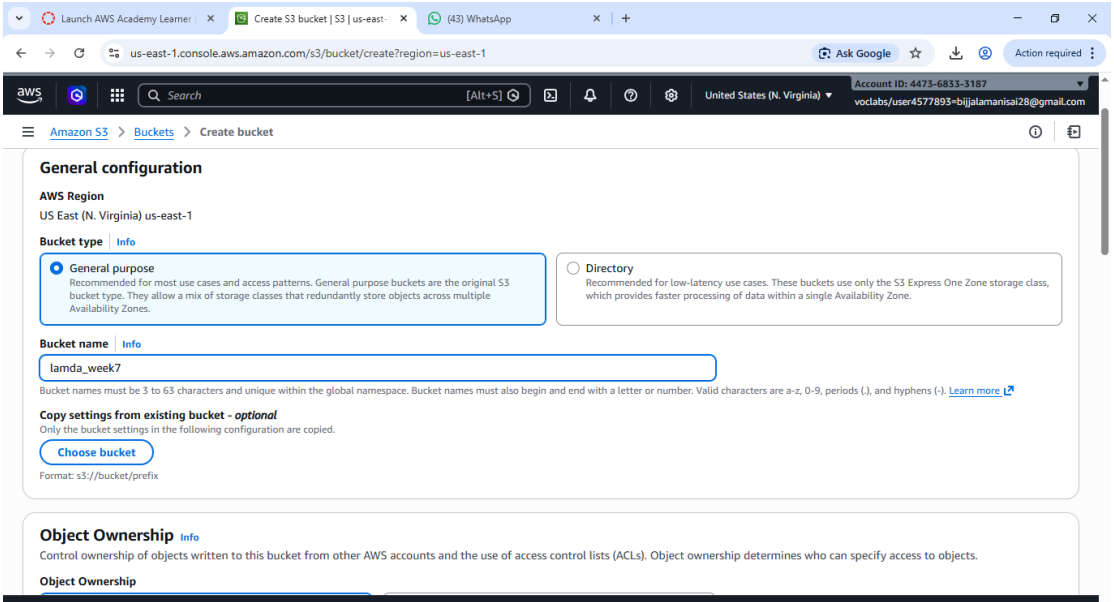
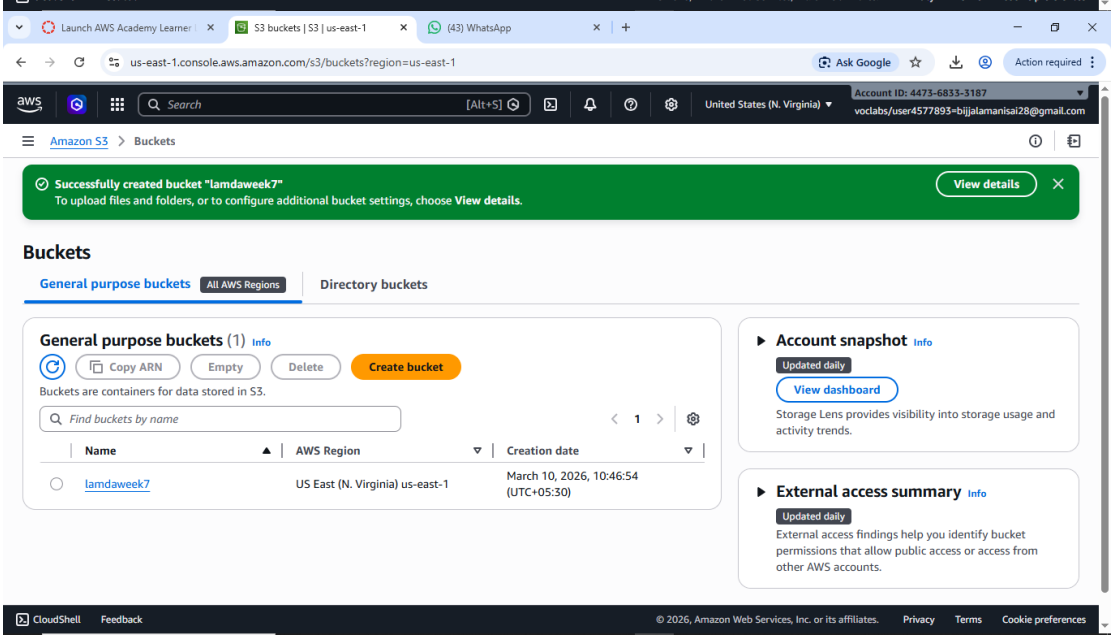
WEEK-7:
step-by-step procedure for setting up an AWS Lambda function to trigger whenever an object is uploaded to an S3 bucket and update a DynamoDB table:

STEP-1: Create the S3 Bucket

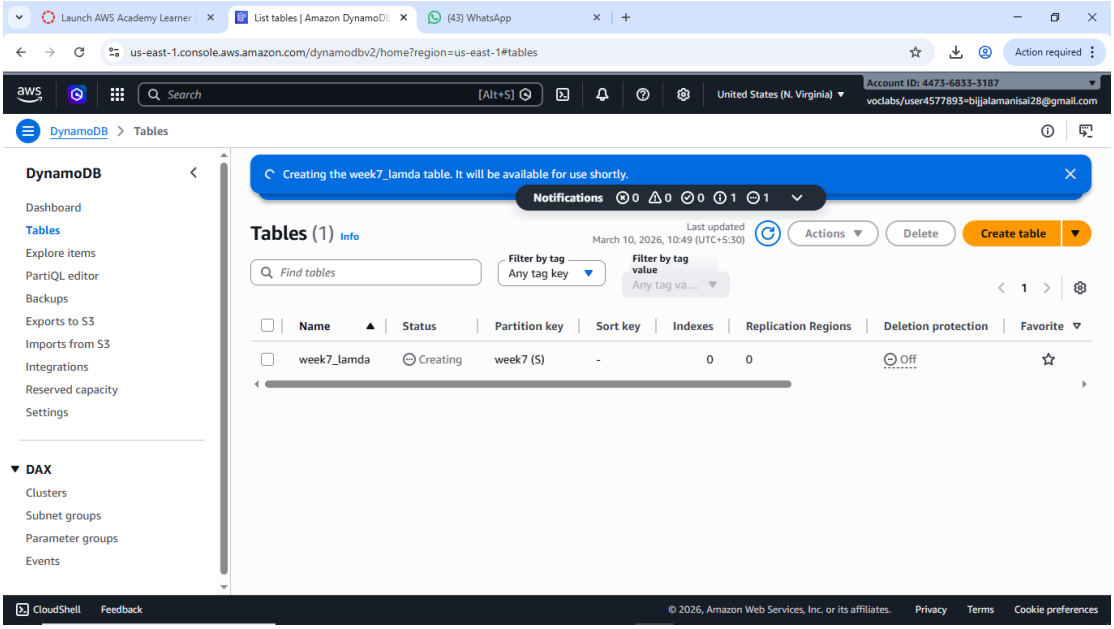
-Search for S3 bucket in amazon console
-Click on the create bucket in the S3 bucket console
-Select general purpose
-Name the bucket as you needed
-Enable Versioning
-Click on the create bucket





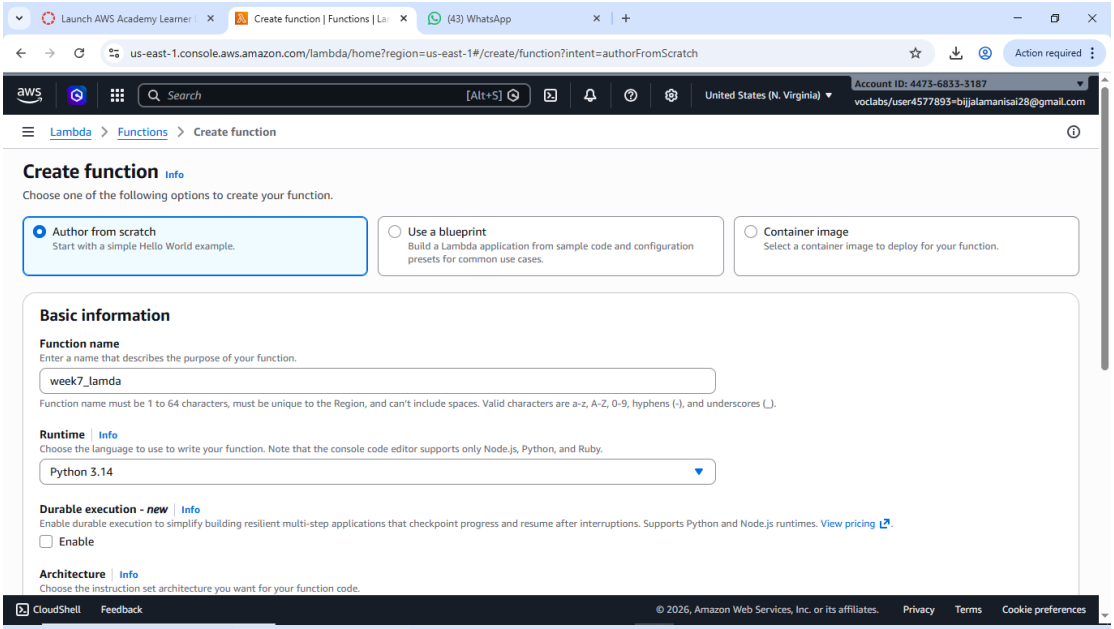
STEP-2: Create a AWS DynamoDB service table

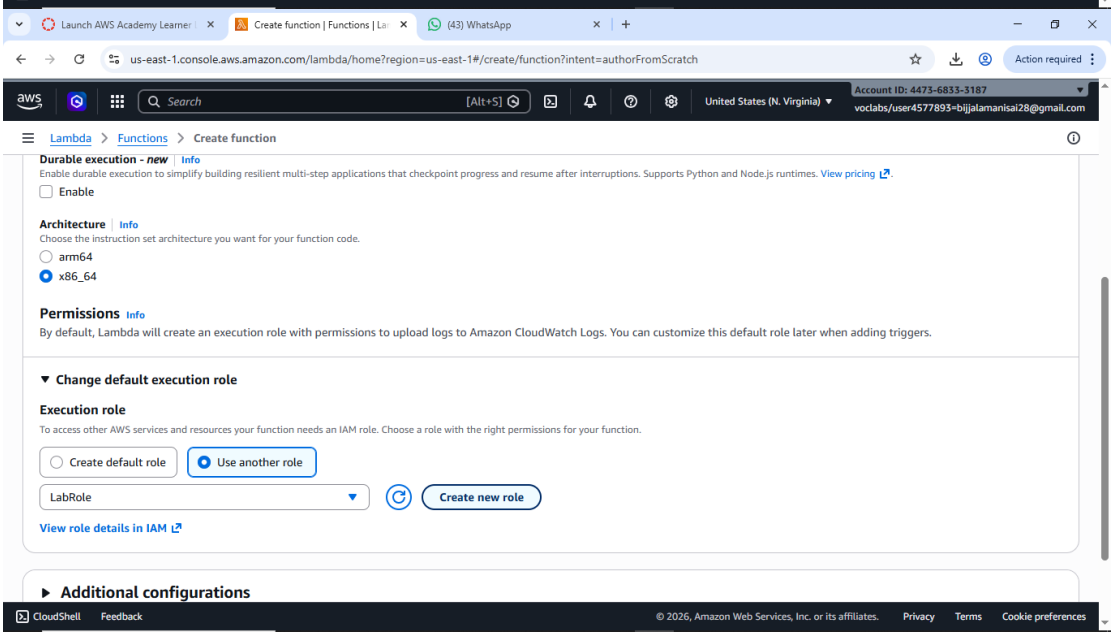
-Search for dynamodb in the search bar of amazon console
-Click on the create table
-GIve any name to the table
Eg. newtable
-Name the row as unique with type string



STEP-3: Creating the lambda function

-Search for aws lambda
-Click on the create function
-Choose Author from scratch
-Type any name in the function name input
-Choose a runtime
Eg.Python
-In permissions choose the role type
If available labrole in the existing roles
Or
Create a new role with basic lambda permissions
-click on the create function





STEP-4: Adding trigger to connect the S3 bucket and DynamoDB

-Click on the Add trigger in the created function
-Search for S3 bucket in the trigger configuration
-Select the S3 bucket
-Choose the bucket which you want to apply the trigger
-Select All object create event in the event type, Acknoledge
-click on the add

Launch AWS Academy Learner | week7_lamda | Functions | Lam | (43) WhatsApp

us-east-1.console.aws.amazon.com/lambda/home?region=us-east-1#/functions/week7_lamda?newFunction=true&tab=code

Account ID: 4473-6833-3187
voclabs/user4577893=bijjalamanisai28@gmail.com

Lambda > Functions > week7_lamda

Successfully created the function **week7_lamda**. You can now change its code and configuration. To invoke your function with a test event, choose "Test".

week7_lamda

Throttle | Copy ARN | Actions

Function overview Info

Export to Infrastructure Composer | Download

Diagram | Template

week7_lamda
Layers (0)

+ Add trigger

+ Add destination

Description
-

Last modified
9 seconds ago

Function ARN
arn:aws:lambda:us-east-1:447368333187:function:week7_lamda

Function URL Info
-

Code | Test | Monitor | Configuration | Aliases | Versions

https://us-east-1.console.aws.amazon.com/lambda/home?region=us-east-1#/add/r...

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us-east-1.console.aws.amazon.com/lambda/home?region=us-east-1#/add/relation?focus=aws%2Flambda&target=arn%3Aaws%3Alambda%3Aus-east...

Account ID: 4473-6833-3187
voclabs/user4577893=bijjalamanisai28@gmail.com

Lambda > Add triggers

Add trigger

Trigger configuration Info

S3
aws asynchronous storage

Bucket
Choose or enter the ARN of an S3 bucket that serves as the event source. The bucket must be in the same region as the function.

s3/lamdaweek7

Bucket region: us-east-1

Event types
Select the events that you want to have trigger the Lambda function. You can optionally set up a prefix or suffix for an event. However, for each bucket, individual events cannot have multiple configurations with overlapping prefixes or suffixes that could match the same object key.

All object create events

Prefix - *optional*
Enter a single optional prefix to limit the notifications to objects with keys that start with matching characters. Any special characters must be URL encoded.

e.g. images/

Suffix - *optional*

CloudShell | Feedback

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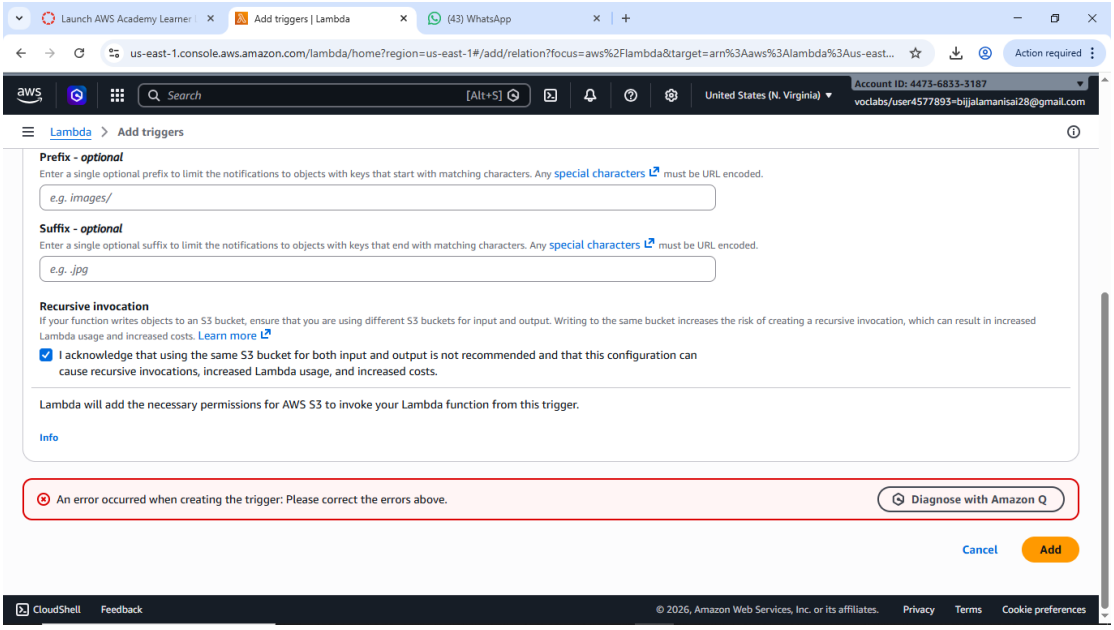
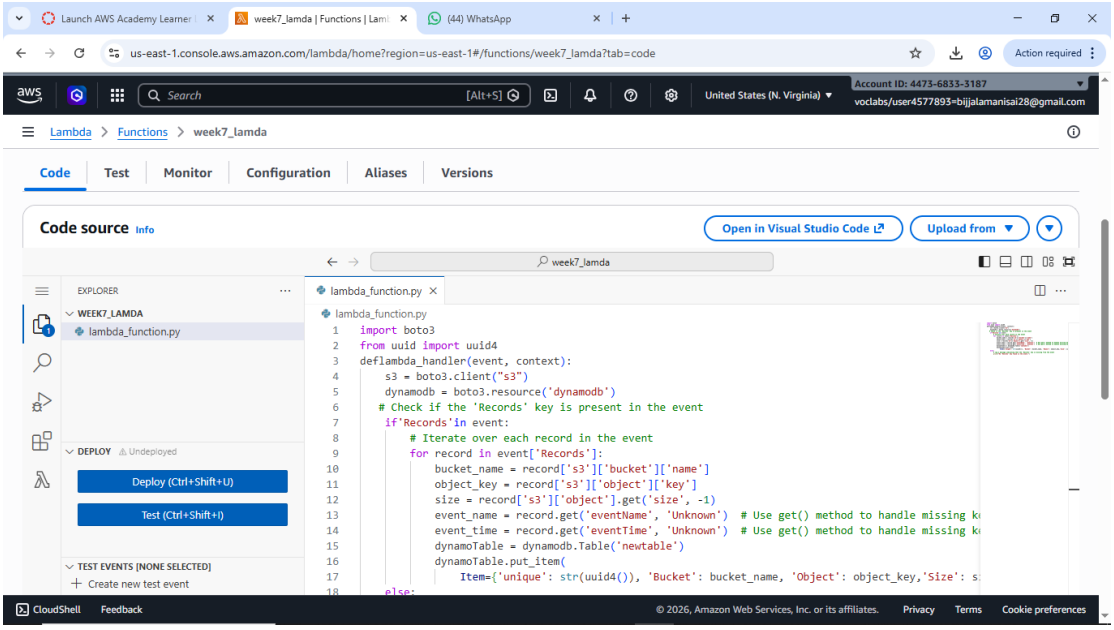


STEP: 5 Creating the deplyoment code and monitoring

-Add the code for triggering when object is uploaded
Eg

```
import boto3
from uuid import uuid4
deflambda_handler(event, context):
    s3 = boto3.client("s3")
    dynamodb = boto3.resource('dynamodb')
   # Check if the 'Records' key is present in the event
    if'Records'in event:
        # Iterate over each record in the event
        for record in event['Records']:
            bucket_name = record['s3']['bucket']['name']
            object_key = record['s3']['object']['key']
            size = record['s3']['object'].get('size', -1)
            event_name = record.get('eventName', 'Unknown')  # Use get() method to
handle missing keys
            event_time = record.get('eventTime', 'Unknown')  # Use get() method to
handle missing keys
            dynamoTable = dynamodb.Table('newtable')
            dynamoTable.put_item(
                Item={'unique': str(uuid4()), 'Bucket': bucket_name, 'Object':
object_key,'Size': size, 'Event': event_name, 'EventTime': event_time})
    else:
        # Log a message indicating that the 'Records' key is missing from the event
        print("No 'Records' key found in the event.")
```



>here in code change the newtable name to dynomo db table name
>in place of unique keep the partion key name
>deploy then

Create a policy if you are getting the Iam role problem Otherwise skip this

-Search for IAM console in AWS console
-Go to roles
-Select the Lamda function name like you gave in the labmda creation

-In permission policies click on add permissions
-Select create inline policy

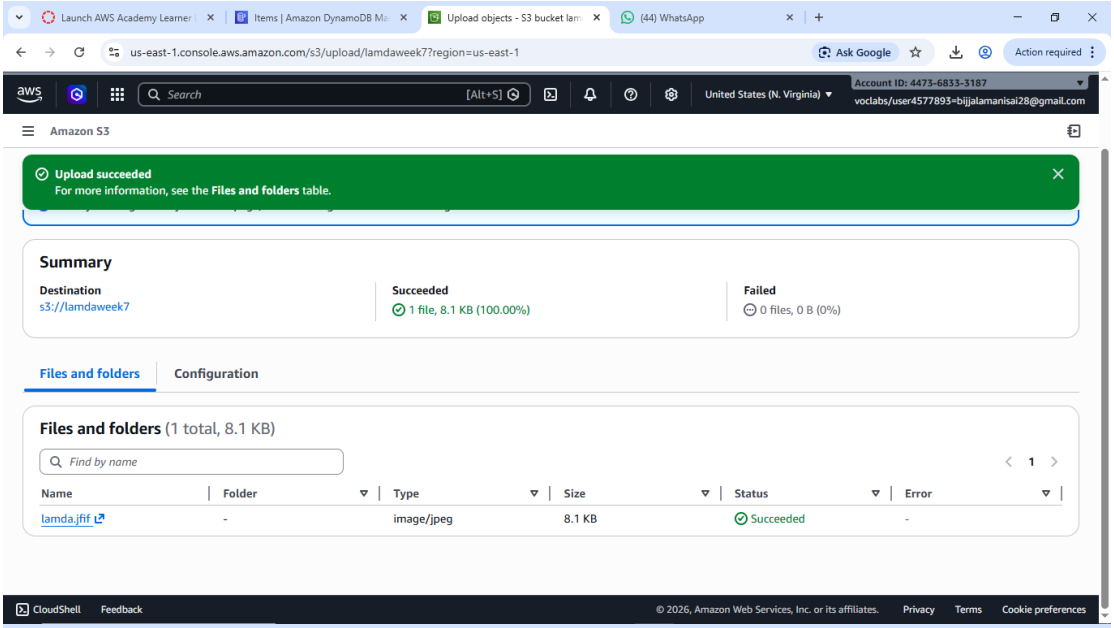
-Click on the JSON in the policy editor
Write this policy code in that editor

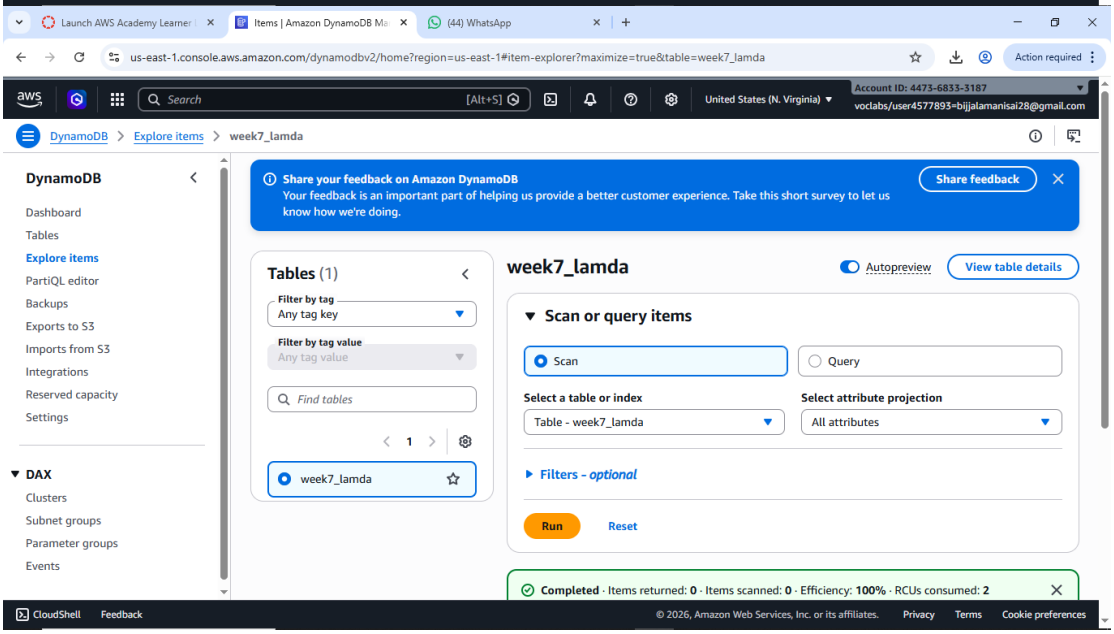
```
{
 "Version": "2012-10-17",
 "Statement": [
  {
   "Effect": "Allow",     "Action": "dynamodb:PutItem",
   "Resource": "arn:aws:dynamodb:us-west-2:669420739378:table/newtable"
  }
 ]
}
```

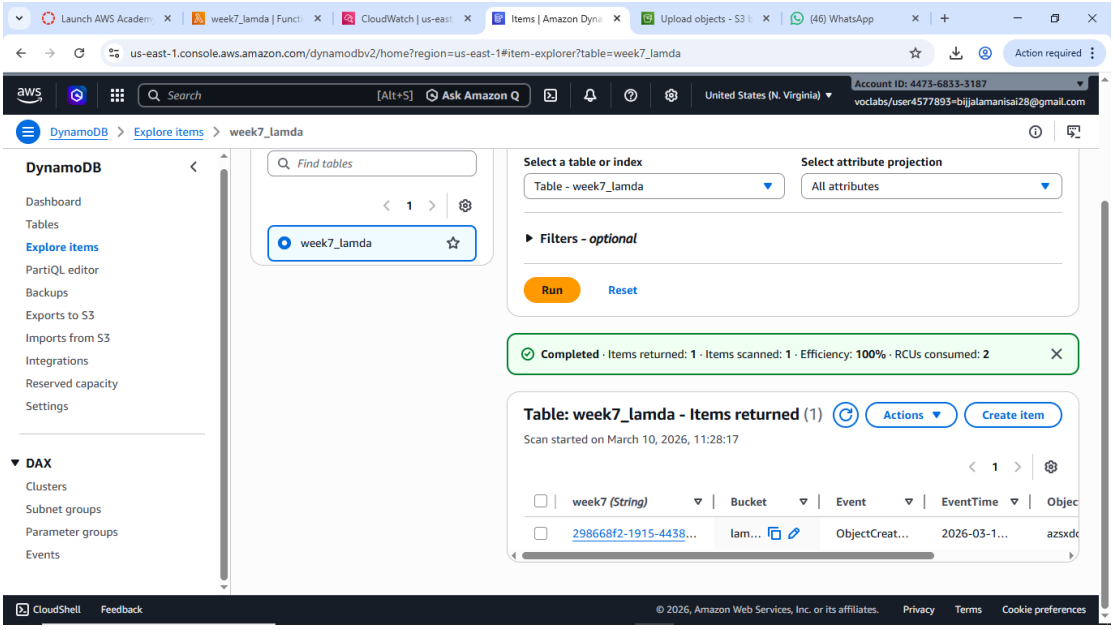
-Click on the next
-Give any policy name
Eg.lambda-dynamo-policy
-Click on the create policy

STEP-6 : Upload any file(s) in the S3 bucket it will triggers and shows in both S3 bucket and the DynamoDB

-Click on the explore items in the DynamoDB
-click on the run
You'll able to see the objects in the dynamodb which is uploaded in the S3 bucket

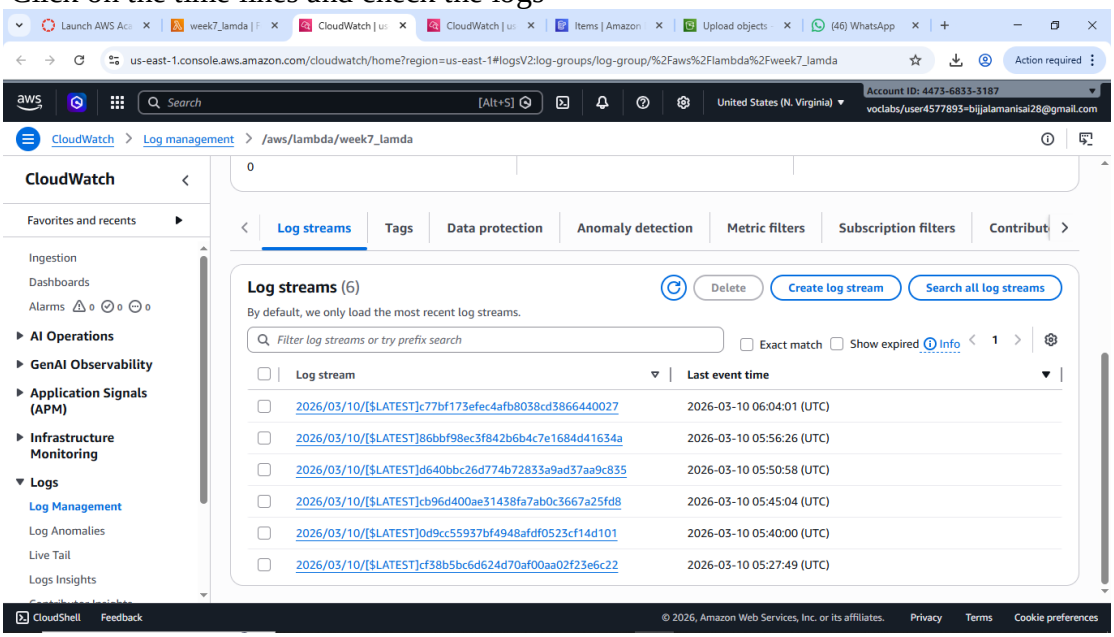


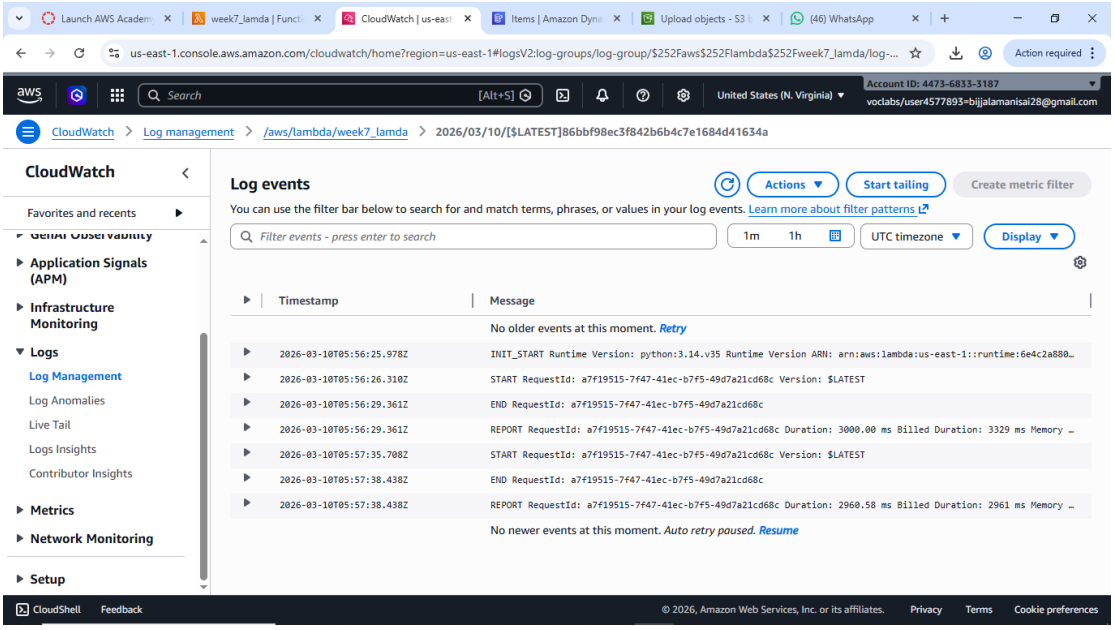




STEP-7: Monitoring using cloud watch

-Click on the Monitor tab in the lambda function
-Click on the view cloudwatch logs
-Click on the time lines and check the logs





SBQ's

1. A company wants to automatically process images whenever they are uploaded to an S3 bucket. The system should resize the images and store them in another bucket. Which AWS service can be used to automatically run the code when the image is uploaded? **Answer:** AWS Lambda can automatically run code when an image is uploaded to Amazon S3.
2. A developer created a Lambda function to store uploaded file details into a DynamoDB table, but the function fails with a permission denied error when writing to the database. What could be the reason and how can it be fixed? **Answer:** The Lambda execution role lacks permission and it can be fixed by adding a policy allowing `dynamodb:PutItem` on Amazon DynamoDB.
3. An application needs to run a piece of code every time a new file is uploaded to an S3 bucket to extract metadata and store it in a database. How would you design this using AWS Lambda? **Answer:** Configure an S3 event trigger to invoke AWS Lambda which extracts metadata and stores it in Amazon DynamoDB.
4. A Lambda function is configured with an S3 trigger, but it is not executing when files are uploaded. What possible configuration issues might cause this problem? **Answer:** Possible issues include incorrect event type, wrong bucket selection, disabled trigger, or missing permissions between Amazon S3 and AWS Lambda.
5. A company wants to process only .jpg image files uploaded to an S3 bucket using Lambda. How can the trigger be configured so that the function runs only for those files? **Answer:** Configure the S3 trigger with a suffix filter `.jpg` in Amazon S3 for the AWS Lambda function.
6. A Lambda function works correctly for a few uploads but fails when hundreds of files are uploaded simultaneously. What AWS feature helps Lambda handle large numbers of events? **Answer:** AWS Lambda automatic scaling handles large numbers of concurrent events.

7. A developer wants to store the table name and bucket name separately from the code so that they can easily change them later. Which Lambda feature can be used to achieve this? **Answer:** Use environment variables in AWS Lambda.
8. A Lambda function executes but does not show any output in the console, making debugging difficult. Which AWS service can be used to view logs and troubleshoot the issue? **Answer:** Use Amazon CloudWatch to view Lambda logs.
9. An organization wants a function to run every day at 10 PM to clean temporary files stored in S3. How can Lambda be triggered automatically at a scheduled time? **Answer:** Use a scheduled rule in Amazon EventBridge to trigger AWS Lambda.
10. A Lambda function processes uploaded files and stores metadata in DynamoDB, but duplicate records appear in the table when the same file is uploaded again. How can this issue be prevented? **Answer:** Use a unique primary key or conditional write in Amazon DynamoDB to avoid duplicates.
11. A Lambda function processes files uploaded to S3, but the processing sometimes takes more than 10 minutes. What limitation of Lambda must be considered here? **Answer:** AWS Lambda has a maximum execution time of 15 minutes.
12. A developer wants to allow a Lambda function to read files from S3 and write records into DynamoDB. What type of AWS security configuration must be created? **Answer:** An IAM execution role with permissions for Amazon S3 and Amazon DynamoDB must be created in AWS Identity and Access Management.
13. A company wants to automatically send a notification email whenever a file is uploaded to S3 using Lambda. Which AWS services can be integrated to achieve this? **Answer:** Integrate Amazon S3, AWS Lambda, and Amazon Simple Notification Service.
14. A Lambda function is triggered by S3 uploads, but the function keeps running multiple times for the same file. What could be the possible cause? **Answer:** Multiple event notifications or retries from Amazon S3 can trigger AWS Lambda multiple times.
15. An organization wants to build a fully serverless application where uploaded files are processed automatically without managing servers. Which AWS services can be combined with Lambda to achieve this architecture? **Answer:** Combine Amazon S3, AWS Lambda, and Amazon DynamoDB for a serverless architecture.